Compressor off-design report: summary and diagnostics

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Rotor-Stator  |  run: Max_eta_Compressor_design_rs_with_last_stator

Rotor-Stator  |  PR_des = 1.4824, eta_des = 0.9123  |  correlations: polynomial, Howell factor 2.00

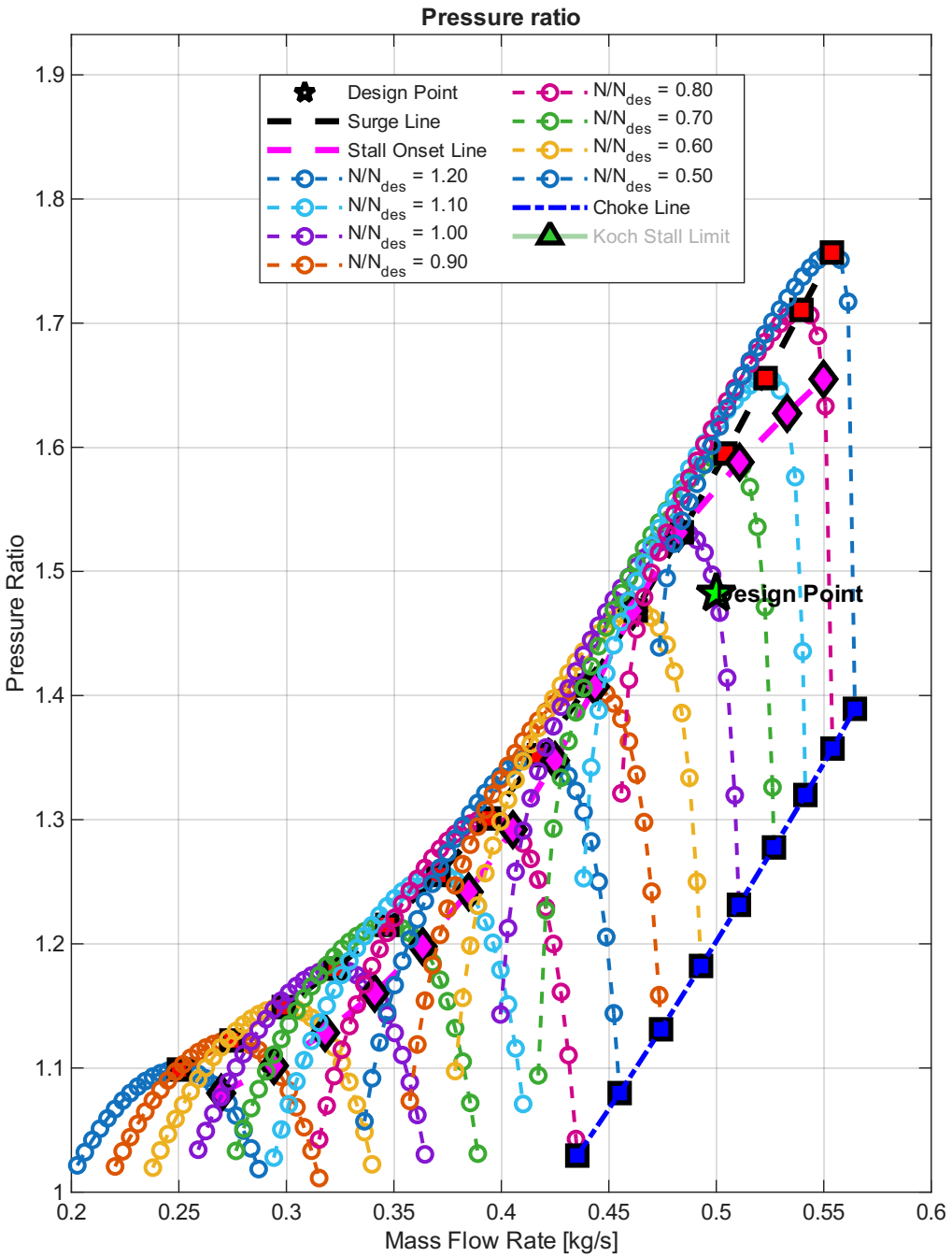
STABILITY (design speed): first limit = peak at m = 0.4840 kg/s (PR 1.532); stall margin SM = 6.7 %
   peak of the speed line (zero slope): m = 0.4840 kg/s | Howell stall onset: m = 0.4825 kg/s
   Koch stage limit: m = 0.5018 kg/s (indicative, see the Koch check below)
CHOKE (design speed): m_choke = 0.5106 kg/s (Stator 2 exit (absolute)) -> choke margin 2.1 %, operating range (m_choke - m_lim)/m_choke = 5.2 %
WARNING: N/N_des = 1.20 has no stable range (none (beyond onset)).
SURGE OR ROTATING STALL (design speed): U/(2a_p) = 0.377 -> surge if the installation has V_p/(A_c L_c) > 4.51 (B > 0.80), rotating stall below.
BLADE ROWS at the first limit (peak): highest loss rise Stator 2 (omega/omega_des = 1.92), lowest de Haller Stator 2 (0.79). Details: Tables > Blade
      rows.

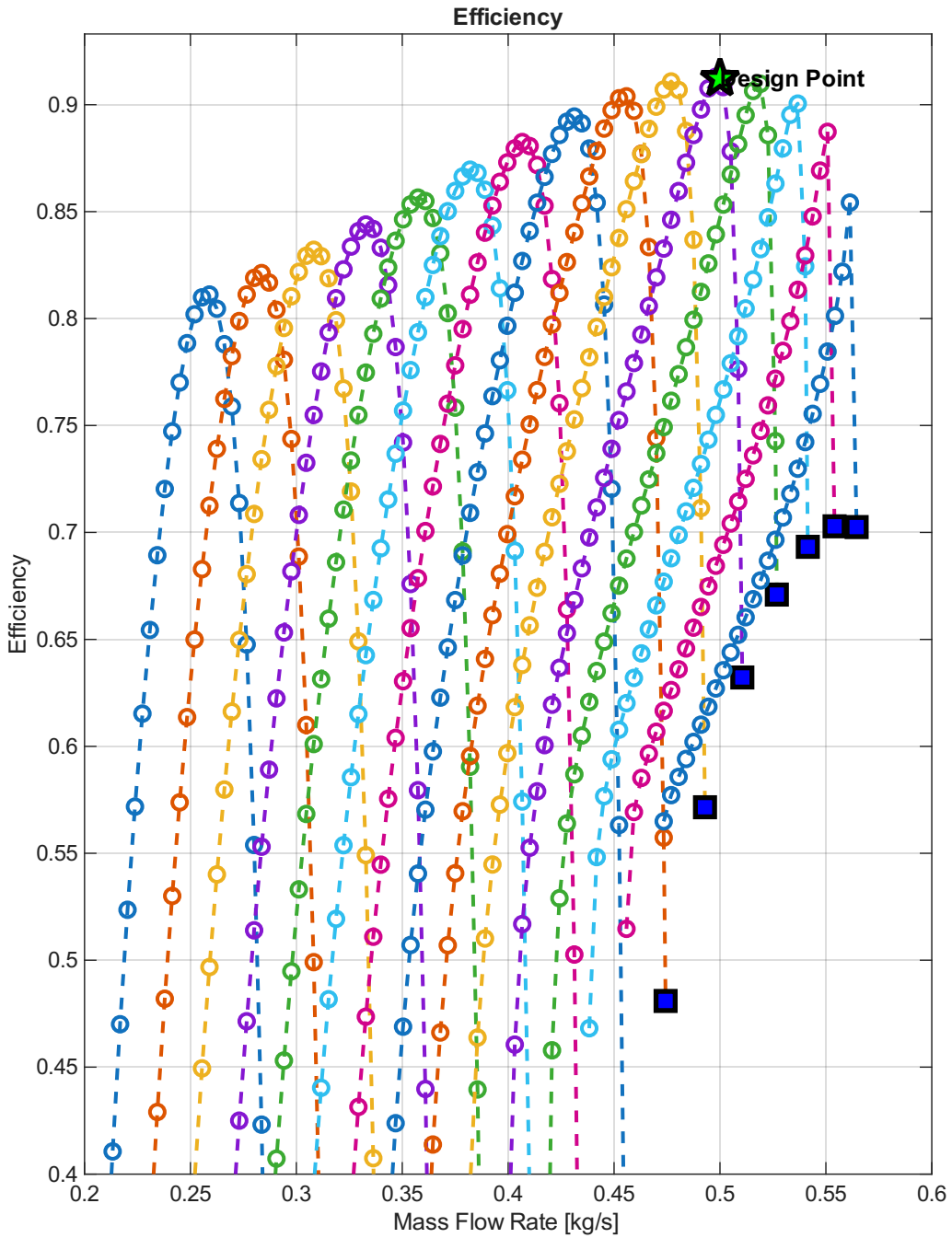
KOCH STAGE-LOADING INDICATOR (never a stability limit): INDICATIVE
   Indicative: near Koch's data. Read the ratio as a trend (more or less loaded than a typical stage), not as a limit. Howell and the peak set the
      limit.
   Koch line: drawn on 6 of 15 speed lines (N/N_des 0.85 to 1.10).
   why: Stage 1 (R1+S1): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
   why: Stage 2 (R2+S2): L/g2 = 0.53 is below 0.6 (low solidity), outside Koch's data, so its stalling pressure rise is extrapolated.
Koch check at the design point (target Ch_ef/Ch_max <= 0.85):
  Stage 1 (R1+S1): BEYOND THE LIMIT AT DESIGN (1.04)
     - RAISE the solidity x1.42 (stage L/g2 0.53 -> 0.76): R1 0.50 -> 0.71, S1 0.50 -> 0.71; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
  Stage 2 (R2+S2): BEYOND THE LIMIT AT DESIGN (1.03)
     - RAISE the solidity x1.41 (stage L/g2 0.53 -> 0.75): R2 0.50 -> 0.70, S2 0.50 -> 0.70; re-run the design script with it (profile loss changes)
     - or LOWER the stage static-pressure rise by about 18% at the same velocities (lower loading psi or degree of reaction, or a larger flow
       coefficient)
     - REDUCE the tip clearance to eps/g = 0.020 (now 0.055) -> ratio 0.95
     - REDUCE the axial gap to dz/s = 0.10 (now 0.38) -> ratio 0.97
     - note: outside Koch's data: L/g2 = 0.53 (data 0.6-2.2) -> indicative only
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Inputs of this run (.mat data and settings)

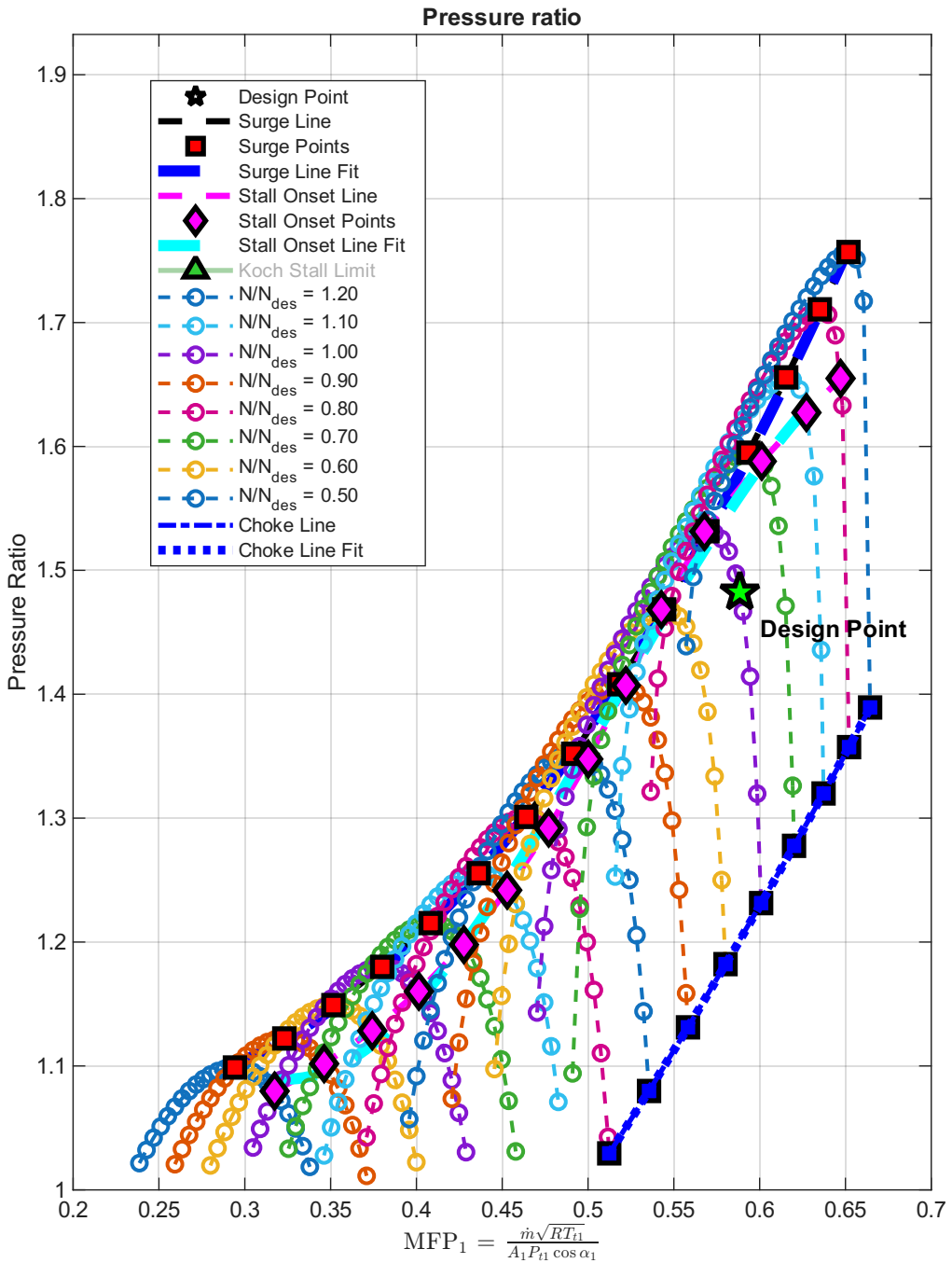
Variable	Value	Unit	Source
Configuration	Rotor-Stator		
Input source	.mat: Max_eta_Compressor_design_rs_...		
alpha1	30.3676	deg	.mat: Max_eta_Compressor_design_rs_...
n_stages	2	-	.mat: Max_eta_Compressor_design_rs_...
A_vec	0.00283128 0.00262358 0.00243314 0....	m^2	.mat: Max_eta_Compressor_design_rs_...
beta1_dash	50.3693	deg	.mat: Max_eta_Compressor_design_rs_...
beta2_dash	18.077	deg	.mat: Max_eta_Compressor_design_rs_...
m_dot	0.5	kg/s	.mat: Max_eta_Compressor_design_rs_...
w_loss_rotor	0.0324827	-	.mat: Max_eta_Compressor_design_rs_...
deviation	12.2907	deg	.mat: Max_eta_Compressor_design_rs_...
incidence	-6.04865	deg	.mat: Max_eta_Compressor_design_rs_...
deflection	13.953	deg	.mat: Max_eta_Compressor_design_rs_...
w_loss_stator	0.0266822	-	.mat: Max_eta_Compressor_design_rs_...
U	274.775	m/s	.mat: Max_eta_Compressor_design_rs_...
sigma_rotor	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
sigma_stator	0.500466	-	.mat: Max_eta_Compressor_design_rs_...
chord	0.01	m	.mat: Max_eta_Compressor_design_rs_...
Correlation set	polynomial		Run settings
Howell factor	2	-	Run settings
Design inlet Tt1	288	K	Run settings
Design inlet pt1	100000	Pa	Run settings
Speed lines N/N_des	0.50 to 1.20 (15 lines)	-	Run settings
Mass-flow points per line	129	-	Run settings
Koch target Ch_ef/Ch_max	0.85	-	Run settings
B critical	0.8	-	Run settings
T_ref / p_ref (corrected map)	288.15 / 101325	K / Pa	Run settings
Koch tip clearance eps/g	0.055	-	Run settings
Koch axial gap dz/s	0.38	-	Run settings

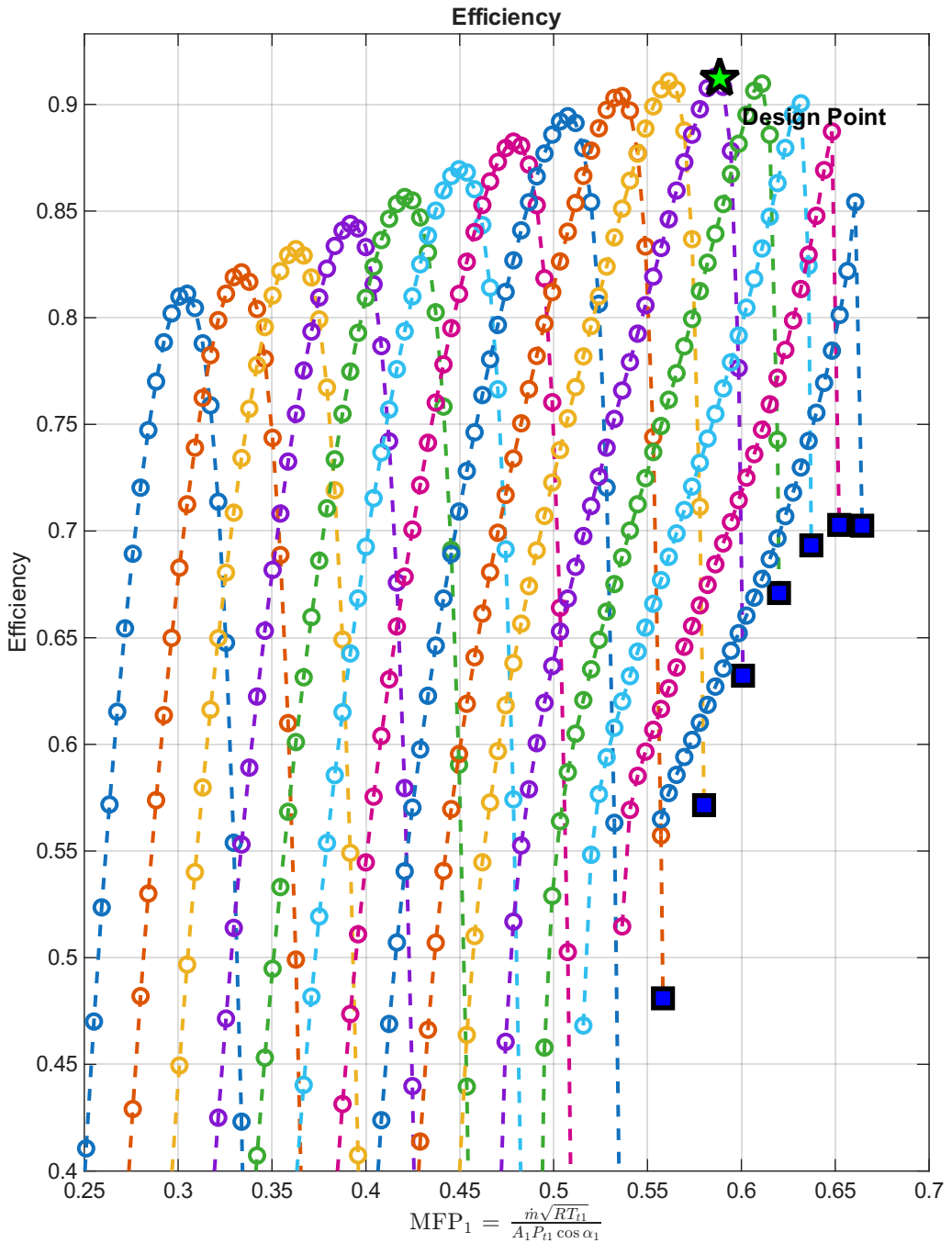
Standard (dimensional) map - Rotor-Stator



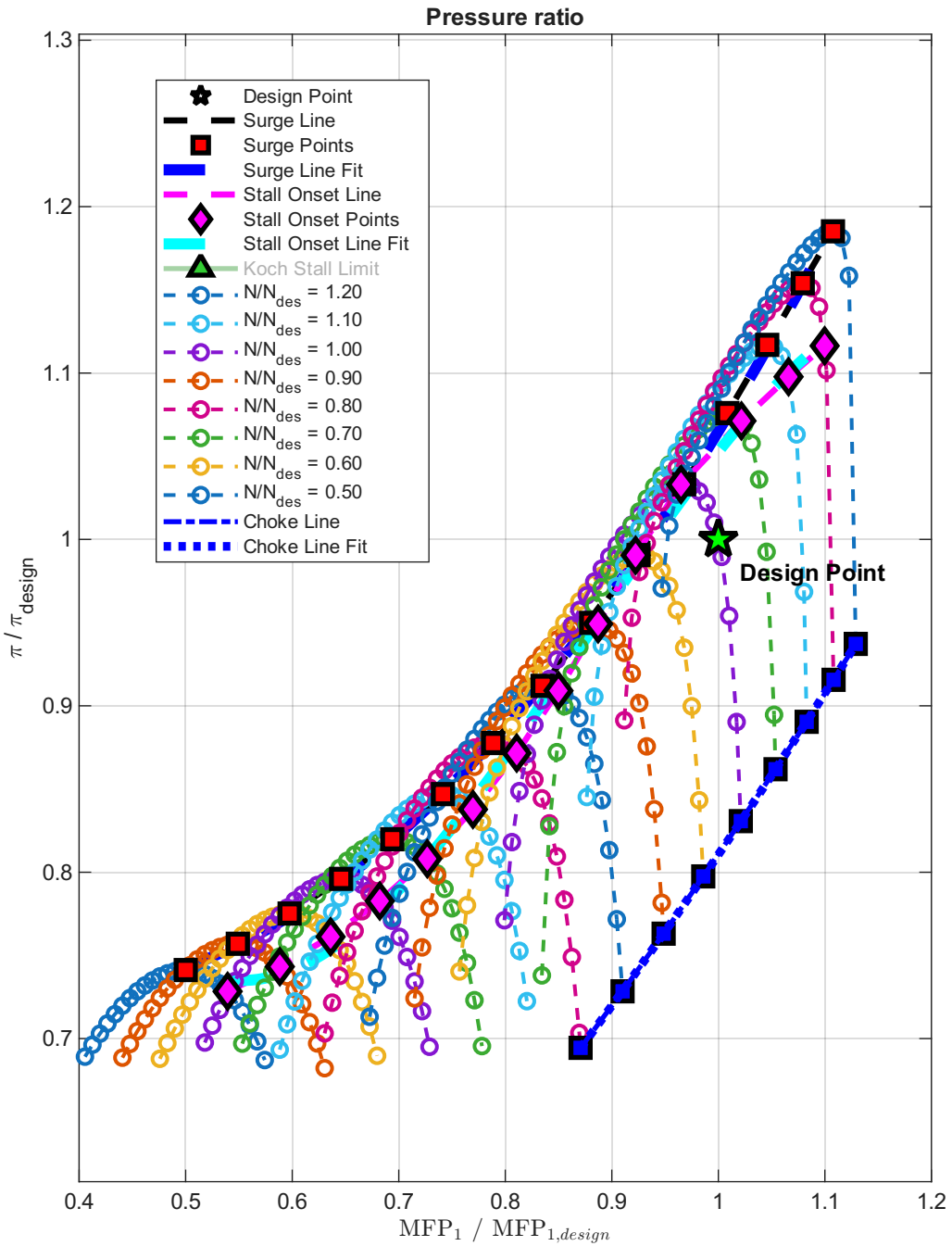


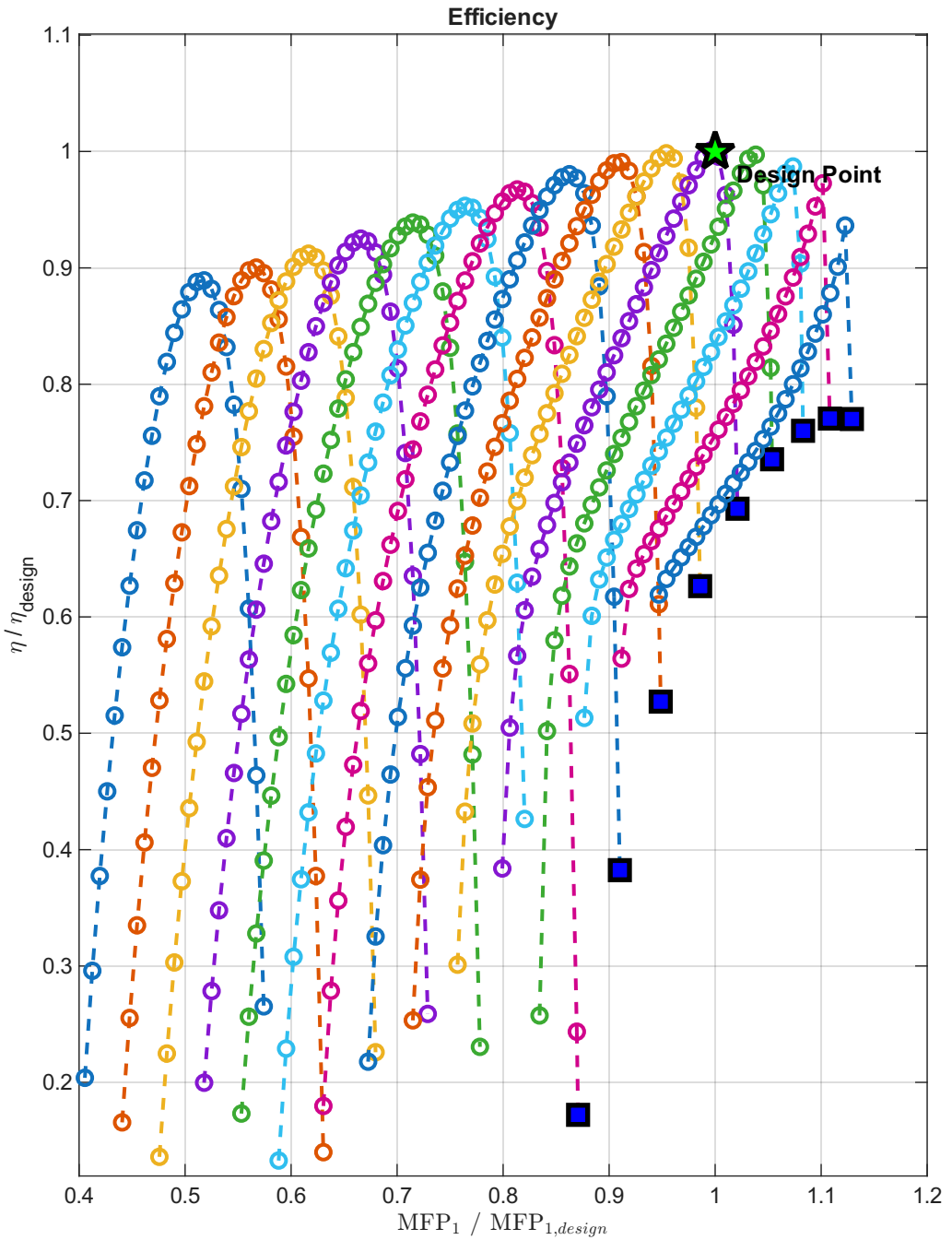
Non-dimensional map - Rotor-Stator



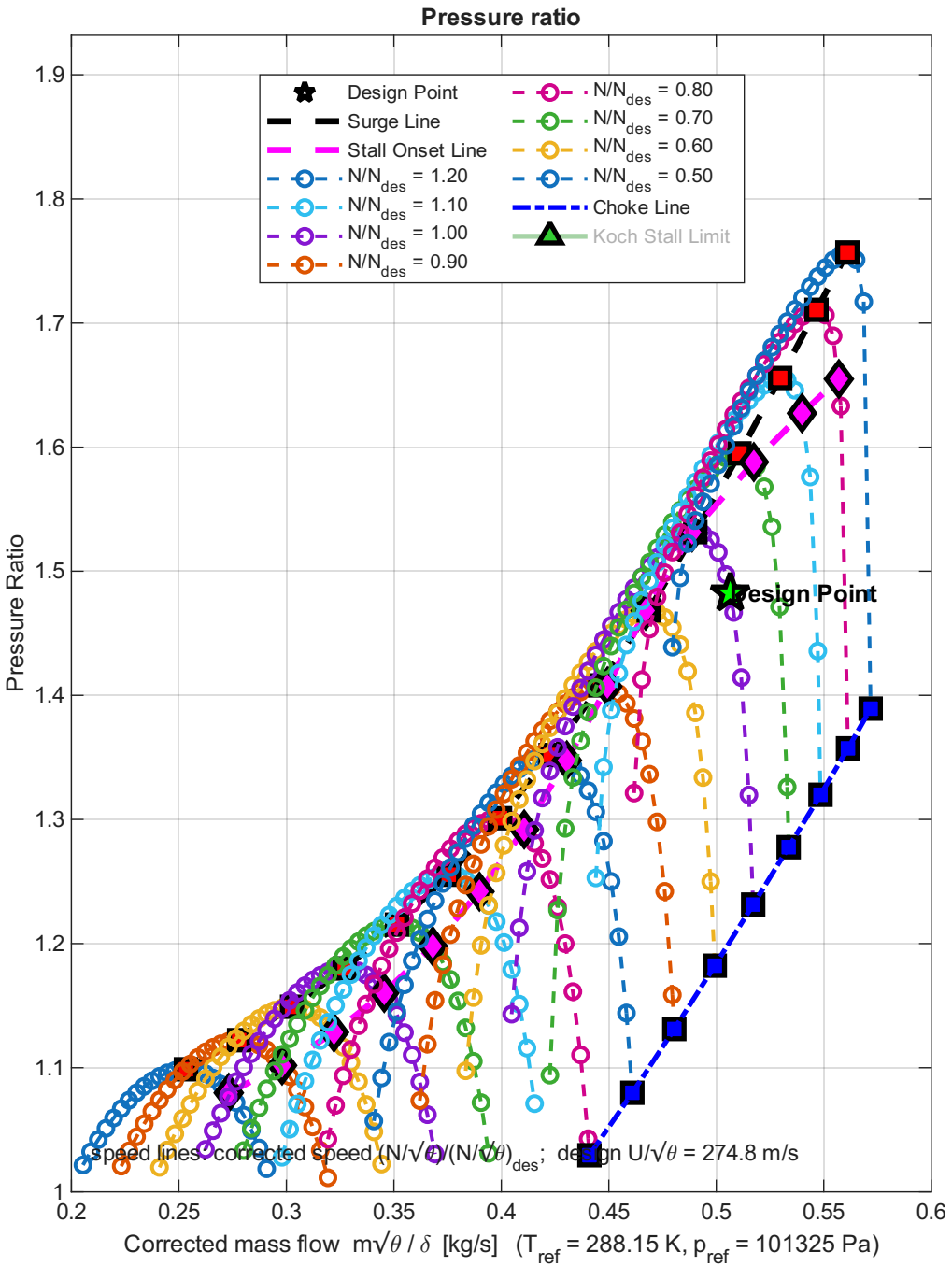


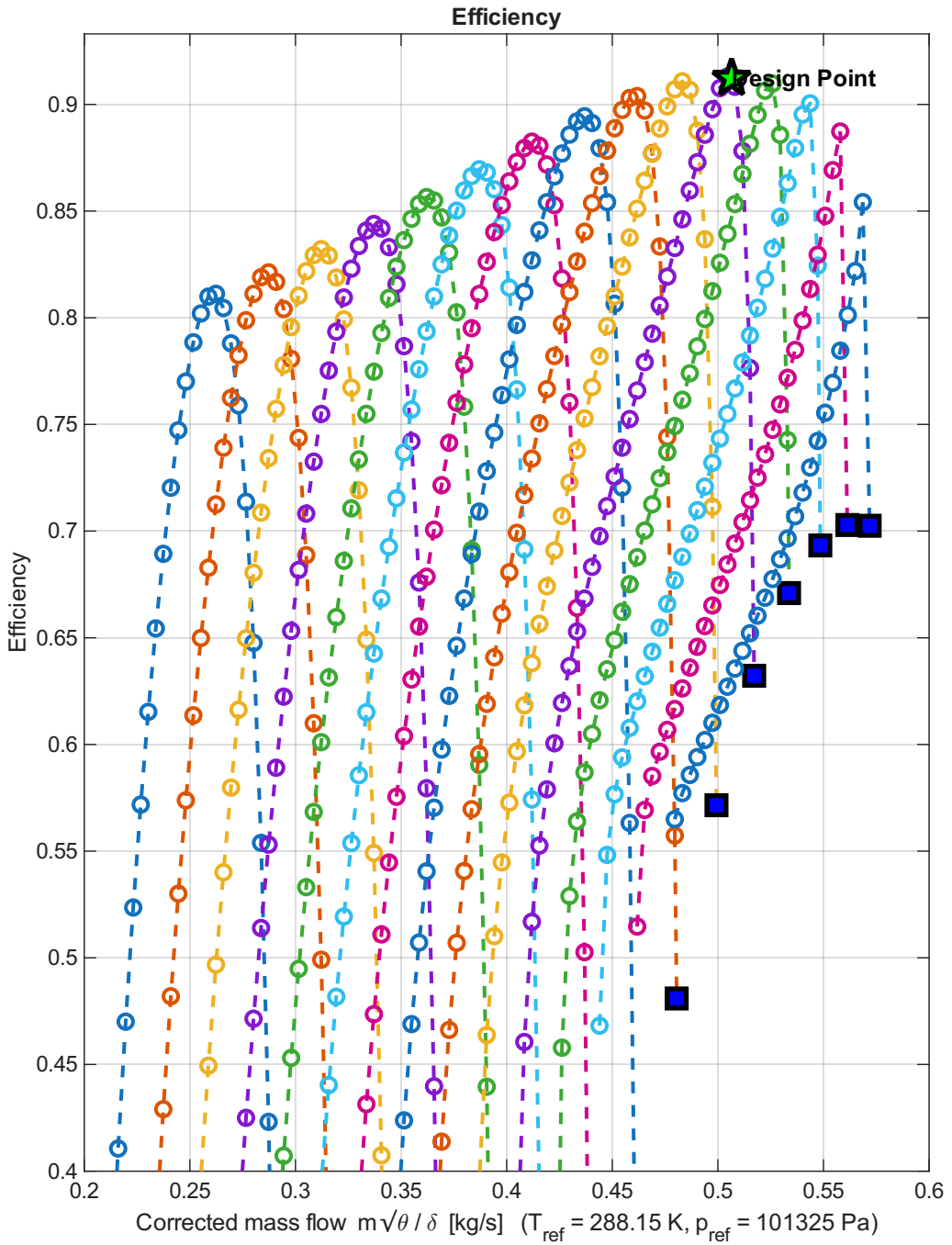
Relative non-dimensional map - Rotor-Stator





Corrected map - Rotor-Stator





Fitted boundary equations (surge, stall onset, choke)

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==== Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
Surge_PR = -0.0025*MFP^3 + 3.4936*MFP^2 + -1.4588*MFP + 1.2279   [R^2 = 0.99991]
Stall-onset line (Howell):
StallOnset_PR = -22.2717*MFP^3 + 34.3856*MFP^2 + -15.2974*MFP + 3.1937   [R^2 = 0.99821]
Choke line:
Choke_PR = 1.3475*MFP^2 + 0.7806*MFP + 0.2755   [R^2 = 0.99999]
==== Relative Non-dimensional Compressor Map: Pressure Ratio - Rotor-Stator ====
Surge line:
(PR/PR_design) = -0.0003*x^3 + 0.8161*x^2 + -0.5791*x + 0.8283   (x = MFP/MFP_design)   [R^2 = 0.99991]
Stall-onset line (Howell):
(PR/PR_design) = -3.0615*x^3 + 8.0324*x^2 + -6.0725*x + 2.1544   (x = MFP/MFP_design)   [R^2 = 0.99821]
Choke line:
(PR/PR_design) = 0.3148*x^2 + 0.3099*x + 0.1859   (x = MFP/MFP_design)   [R^2 = 0.99999]
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Design point

Quantity	Value
Configuration	Rotor-Stator
Pressure ratio PR_des	1.4824
Isentropic efficiency eta_des	0.91226
Mass flow m_des [kg/s]	0.5
MFP_1,des	0.58846
Corrected mass flow [kg/s]	0.50649
Inlet Tt1 [K] / Pt1 [Pa]	288 / 100000
Correlation set	polynomial
Howell factor	2

Choke

N/N_des	m_choke [kg/s]	MFP_1	PR	Eff	Choking row	Inside compressor region (PR>1)
0.5	0.31677	0.3728	0.73		Stator 2 exit (absolute)	no
0.55	0.33614	0.3956	0.778		Stator 2 exit (absolute)	no
0.6	0.35589	0.4188	0.826		Stator 2 exit (absolute)	no
0.65	0.37579	0.4423	0.876		Stator 2 exit (absolute)	no
0.7	0.39565	0.4656	0.927		Stator 2 exit (absolute)	no
0.75	0.41552	0.489	0.978		Stator 2 exit (absolute)	no
0.8	0.43535	0.5124	1.029	0.157	Stator 2 exit (absolute)	yes
0.85	0.45506	0.5356	1.08	0.349	Stator 2 exit (absolute)	yes
0.9	0.47433	0.5582	1.131	0.481	Stator 2 exit (absolute)	yes
0.95	0.49293	0.5801	1.182	0.572	Stator 2 exit (absolute)	yes
1	0.51055	0.6009	1.231	0.632	Stator 2 exit (absolute)	yes
1.05	0.52685	0.6201	1.278	0.671	Stator 2 exit (absolute)	yes
1.1	0.54146	0.6373	1.32	0.693	Stator 2 exit (absolute)	yes
1.15	0.5541	0.6521	1.357	0.703	Stator 2 exit (absolute)	yes
1.2	0.56451	0.6644	1.389	0.703	Stator 2 exit (absolute)	yes

Stall onset (Howell)

N/N_des	m_stall [kg/s]	PR	Eff	Stalling row
0.5	0.26968	1.08	0.757	Rotor 1
0.55	0.2942	1.102	0.78	Rotor 1
0.6	0.31802	1.128	0.804	Rotor 1
0.65	0.34111	1.16	0.828	Rotor 1
0.7	0.36339	1.198	0.85	Rotor 1
0.75	0.38483	1.242	0.869	Rotor 1
0.8	0.40538	1.292	0.882	Rotor 1
0.85	0.42497	1.348	0.887	Rotor 1
0.9	0.44358	1.407	0.884	Rotor 1
0.95	0.46116	1.468	0.871	Rotor 1
1	0.48249	1.531	0.867	Stator 2
1.05	0.51086	1.588	0.89	Stator 2
1.1	0.53298	1.627	0.894	Stator 2
1.15	0.5499	1.655	0.885	Stator 2
1.2	NaN	NaN	NaN	beyond onset on the whole line (Sta...

Peak (zero slope)

N/N_des	m_peak [kg/s]	PR	Eff	m_stall - m_peak [kg/s]	First limit on throttling
0.5	0.25002	1.099	0.795	0.01966	stall onset (Howell)
0.55	0.27447	1.122	0.804	0.01973	stall onset (Howell)
0.6	0.29878	1.149	0.814	0.01924	stall onset (Howell)
0.65	0.32294	1.18	0.825	0.01816	stall onset (Howell)
0.7	0.34694	1.215	0.837	0.01645	stall onset (Howell)
0.75	0.37073	1.255	0.848	0.0141	stall onset (Howell)
0.8	0.39428	1.301	0.858	0.0111	stall onset (Howell)
0.85	0.41751	1.352	0.867	0.00746	stall onset (Howell)
0.9	0.44032	1.408	0.873	0.00326	stall onset (Howell)
0.95	0.46256	1.468	0.876	-0.0014	peak (zero slope)
1	0.48398	1.532	0.873	-0.00149	peak (zero slope)
1.05	0.50426	1.595	0.864	0.0066	stall onset (Howell)
1.1	0.52296	1.656	0.849	0.01002	stall onset (Howell)
1.15	0.53963	1.711	0.827	0.01028	stall onset (Howell)
1.2	0.55382	1.757	0.799	NaN	-

Koch limit

N/N_des	m_Koch [kg/s]	PR	Eff	Limiting stage
0.5	NaN	NaN	NaN	not reached
0.55	NaN	NaN	NaN	not reached
0.6	NaN	NaN	NaN	not reached
0.65	NaN	NaN	NaN	not reached
0.7	NaN	NaN	NaN	not reached
0.75	NaN	NaN	NaN	not reached
0.8	NaN	NaN	NaN	not reached
0.85	0.42926	1.34	0.894	Stage 1 (R1+S1)
0.9	0.45742	1.374	0.902	Stage 1 (R1+S1)
0.95	0.48085	1.416	0.906	Stage 1 (R1+S1)
1	0.50176	1.465	0.907	Stage 1 (R1+S1)
1.05	0.52159	1.496	0.899	Stage 2 (R2+S2)
1.1	0.53879	1.513	0.879	Stage 2 (R2+S2)
1.15	NaN	NaN	NaN	beyond the limit on the whole line ...
1.2	NaN	NaN	NaN	beyond the limit on the whole line ...

Stability limit & Greitzer B

N/N_des	First limit	m_lim [kg/s]	PR	U [m/s]	a_p [m/s]	U/(2a_p)	(V_p/(A_c L_c))_crit	B	Beyond the limit
0.5	stall onset	0.26968	1.08	137.4	345.1	0.199	16.15	NaN	-
0.55	stall onset	0.2942	1.102	151.1	346.2	0.2182	13.44	NaN	-
0.6	stall onset	0.31802	1.128	164.9	347.5	0.2372	11.37	NaN	-
0.65	stall onset	0.34111	1.16	178.6	349	0.2559	9.77	NaN	-
0.7	stall onset	0.36339	1.198	192.3	350.6	0.2743	8.51	NaN	-
0.75	stall onset	0.38483	1.242	206.1	352.5	0.2924	7.49	NaN	-
0.8	stall onset	0.40538	1.292	219.8	354.5	0.31	6.66	NaN	-
0.85	stall onset	0.42497	1.348	233.6	356.8	0.3273	5.98	NaN	-
0.9	stall onset	0.44358	1.407	247.3	359.4	0.3441	5.41	NaN	-
0.95	peak	0.46256	1.468	261	362	0.3605	4.92	NaN	-
1	peak	0.48398	1.532	274.8	364.5	0.3769	4.51	NaN	-
1.05	stall onset	0.51086	1.588	288.5	366.2	0.394	4.12	NaN	-
1.1	stall onset	0.53298	1.627	302.3	367.5	0.4113	3.78	NaN	-
1.15	stall onset	0.5499	1.655	316	368.7	0.4285	3.49	NaN	-
1.2	none (beyond onset)	NaN	NaN	329.7	NaN	NaN	NaN	NaN	-

Blade rows

Operating point (N = 1)	m [kg/s]	Blade row	Incidence i [deg]	(i - i_des)/deflection	Loss omega	omega/omega_des	M_in (row frame)	M_out (row frame)	de Haller V_out/V_in	Deviation [deg]	Watch
Design point	0.5	Rotor 1	-6.05	0	0.0325	1	0.75	0.611	0.829	12.29	
Design point	0.5	Stator 1	-6.05	0	0.0267	1	0.737	0.601	0.829	12.29	
Design point	0.5	Rotor 2	-6.05	0	0.0325	1	0.725	0.592	0.829	12.29	
Design point	0.5	Stator 2	-6.05	0	0.0267	1	0.713	0.582	0.829	12.29	
First limit (peak)	0.48398	Rotor 1	-3.73	0.166	0.0494	1.522	0.74	0.575	0.793	12.74	
First limit (peak)	0.48398	Stator 1	-3.65	0.172	0.0449	1.681	0.72	0.561	0.794	12.76	near Howell
First limit (peak)	0.48398	Rotor 2	-3.42	0.188	0.0543	1.672	0.706	0.549	0.792	12.84	near Howell
First limit (peak)	0.48398	Stator 2	-3.27	0.199	0.0511	1.917	0.692	0.538	0.791	12.9	near Howell
Near choke (0.99 m_choke)	0.50545	Rotor 1	-6.87	-0.059	0.0347	1.068	0.754	0.629	0.847	12.24	
Near choke (0.99 m_choke)	0.50545	Stator 1	-7.22	-0.084	0.0311	1.165	0.744	0.627	0.855	12.24	
Near choke (0.99 m_choke)	0.50545	Rotor 2	-7.75	-0.122	0.0417	1.283	0.735	0.63	0.868	12.23	
Near choke (0.99 m_choke)	0.50545	Stator 2	-8.59	-0.182	0.0469	1.76	0.729	0.642	0.89	12.25	